ONE SET OF LAWS, TIGHTER BASES

The product, quotient and power-of-a-power rules hold for every exponent type. What changes is which bases are allowed: the restrictions tighten as the exponent type broadens.

LAWS OF EXPONENTS

$$a^m \cdot a^n = a^{m+n}$$

split by **the type of exponent**

$\mathbb{R}$

CASE 1

natural

Any base

≠0

CASE 2

zero or negative

Excludes $a = 0$

≥0

CASE 3

rational, even denominator

Requires $a \geq 0$

>0

CASE 4

irrational

Requires $a > 0$

TAKEAWAY

Each new exponent type is defined so that the same laws stay valid. The exponent rules page has every law with its full domain conditions.